

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

ELECTRONIC MEASUREMENTS AND INSTRUMENTATION

(Electronics and Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Define Accuracy. | 2M |
| (b) List the advantages of series type multimeters. | 2M |
| (c) List the applications of CRO. | 2M |
| (d) What is the function of digital frequency counter? | 2M |
| (e) Mention the applications of random noise generator. | 2M |
| (f) What is the significance of spectrum analyzer? | 2M |
| (g) List the characteristics of Maxwell's bridge. | 2M |
| (h) How interference can be reduced in bridges? | 2M |
| (i) Distinguish between active and passive transducers. | 2M |
| (j) What is the need for pH measurement? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|---|-----|
| 2 | Explain the different types of errors in measurement systems. | 10M |
| OR | | |
| 3 | Explain in detail about the working principle of DC ammeters with suitable diagram. | 10M |
| 4 | Draw the block diagram of a dual beam oscilloscope and explain its working. | 10M |
| OR | | |
| 5 | Discuss in detail about Digital Storage Oscilloscope (DSO). | 10M |
| 6 | With a neat diagram explain the operation of a pulse generator. | 10M |
| OR | | |
| 7 | Explain in detail about Harmonic distortion analyzer. | 10M |
| 8 | Explain the operation of the Wheatstone bridge with suitable diagrams. | 10M |
| OR | | |
| 9 | Explain the construction and working of Anderson Bridge with suitable diagrams. | 10M |
| 10 | Explain the construction and working of LVDT. | 10M |
| OR | | |
| 11 | Mention the principle of operation of a piezoelectric transducer. Explain any one piezoelectric transducer in detail. | 10M |
